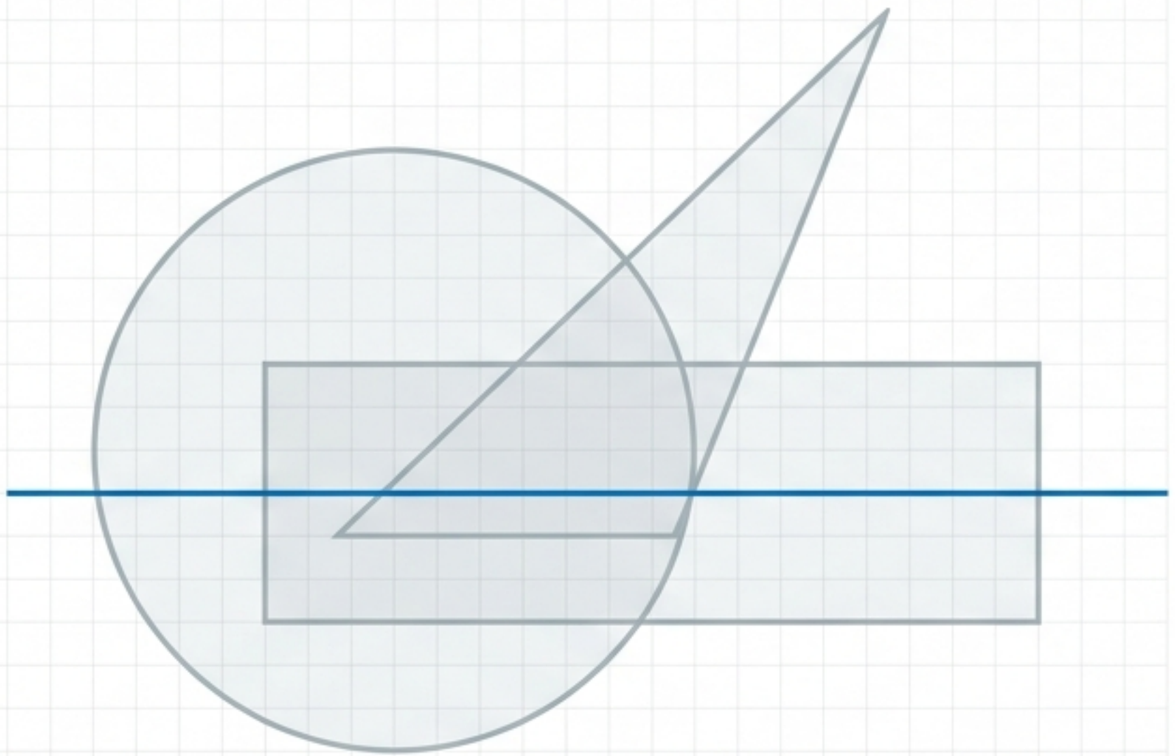
Module 9, Lesson 3

AI 101 FOR MECHANICAL ENGINEERS

Grouping, compression and fair selection



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Lesson Structure

1. Unsupervised Analytical Methods
2. Feature Scaling Implications
3. Model Selection Protocols
4. Technical Interpretation and Limits

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1

1. Unsupervised Analytical Methods

Defining foundational tools for unlabelled data grouping and feature compression.

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Foundational Unsupervised Algorithms



k-means

- Groups data points around selected mathematical centres.
- Minimizes the computed within-cluster distance objective.
- Example: Grouping unlabelled mechanical operating profiles.



PCA

- Projects data vectors onto high-variance directions.
- Compresses highly correlated sensor feature inputs.
- Output: Principal components ranked by captured variance.

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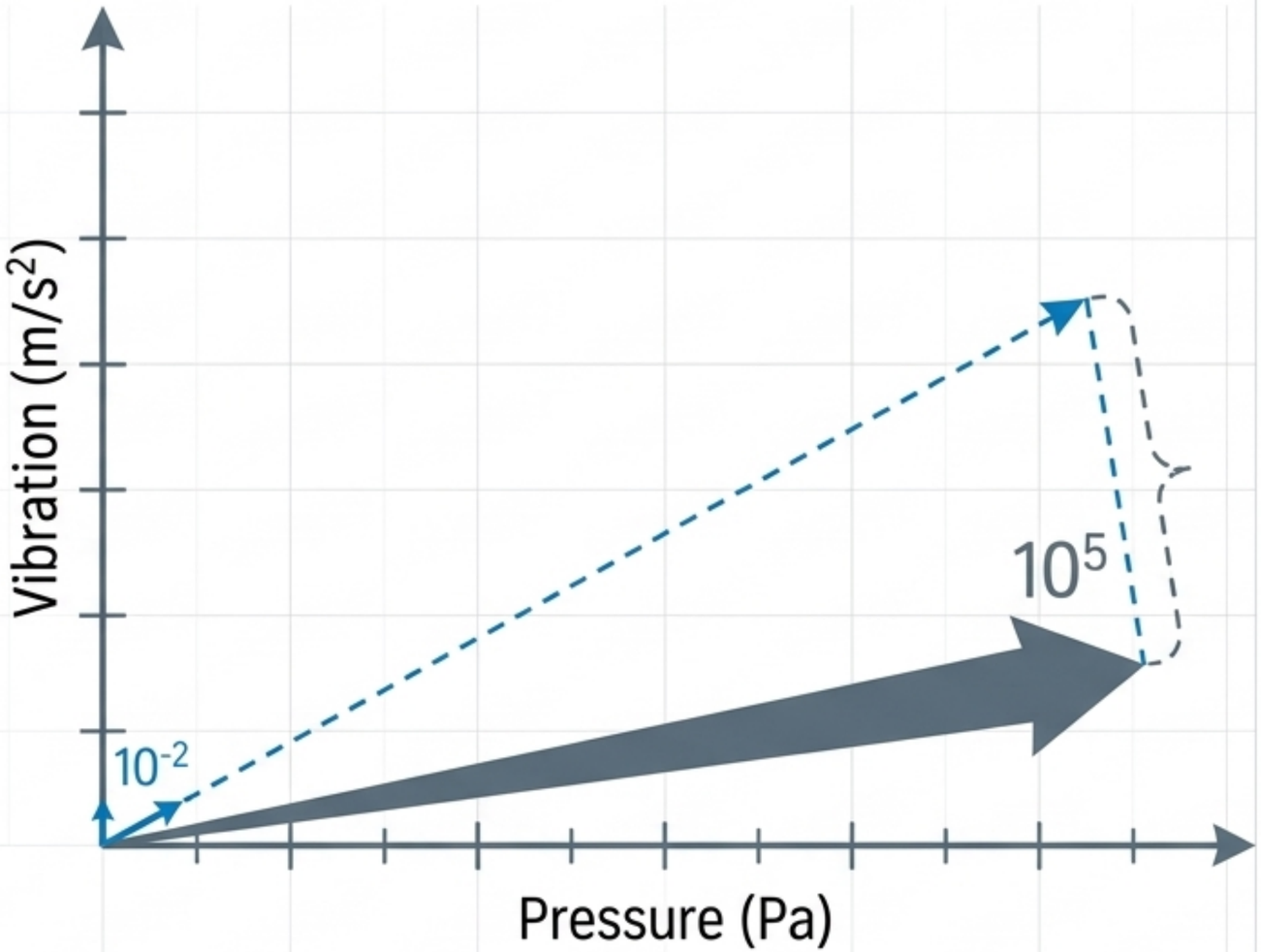
2. Feature Scaling Implications

Addressing algorithmic sensitivity to unit magnitudes in geometric distance calculations.

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Magnitude Dictates Distance Calculations

- Distance-based methods require careful feature scaling.
- Unscaled large-magnitude variables dominate geometric distance computations.
- Crucial small-magnitude vibration features become mathematically negligible.
- Select preprocessing methods matching physical task units.



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3. Model Selection Protocols

Establishing strict evaluation frameworks for candidate classification models.

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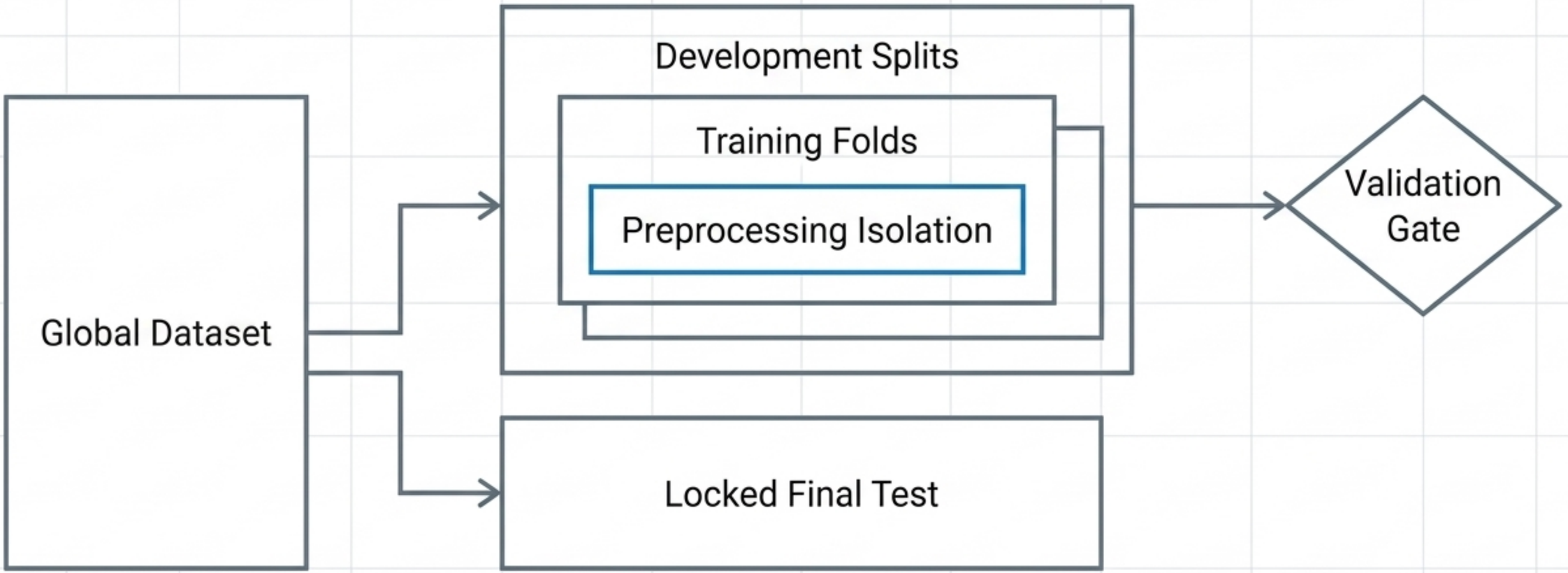
Classification Candidate Selection

Candidate Method	Engineering Characteristic
Logistic Regression	Provides a baseline linear classification boundary.
Decision Trees	Generates highly inspectable logical partition rules.
k-Nearest Neighbours	Utilizes scaled, distance-based similarity metrics.
Ensembles	Requires proven performance to justify added complexity.

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Rigorous Validation Pipeline Architecture



- Compare candidate models on common development data.

- Fit feature preprocessing exclusively inside training folds.

- Select optimal candidates using separate validation sets.

- Evaluate the final locked choice on isolated test data.

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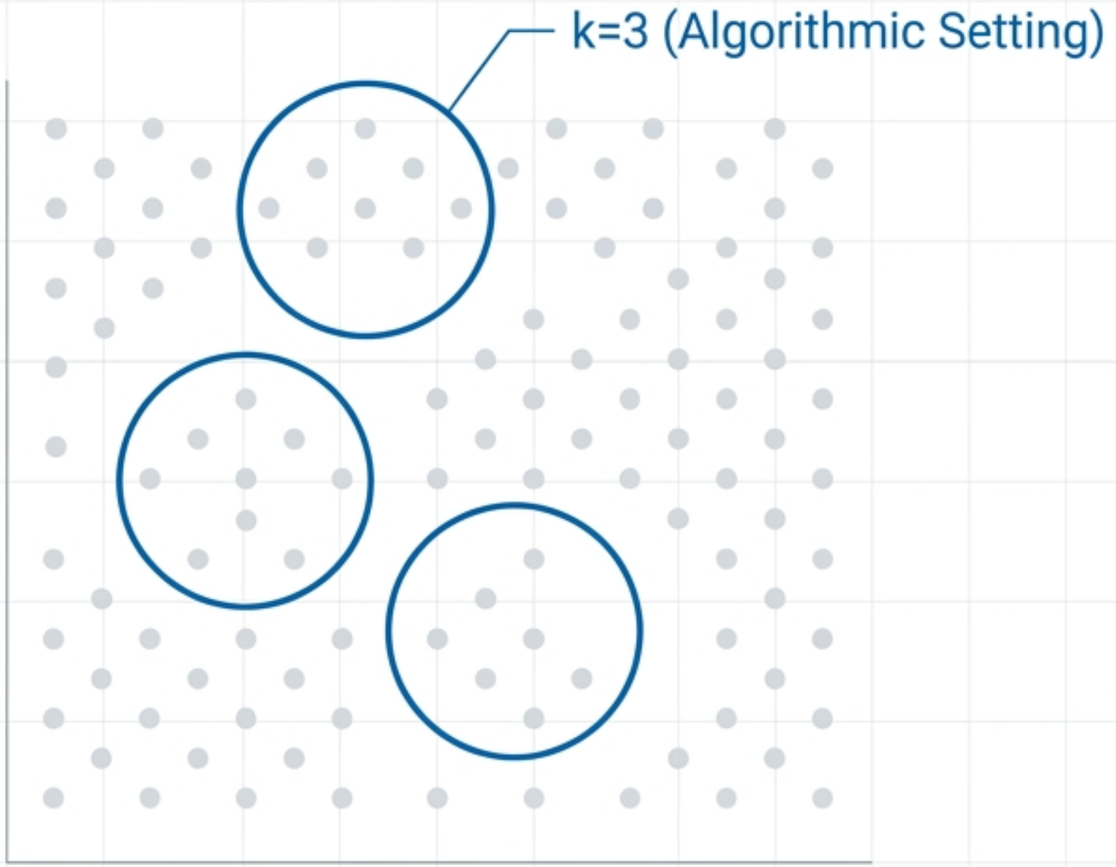
4. Technical Interpretation and Limits

Synthesizing machine learning outputs with physical engineering realities.

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k-means Interpretation Limitations

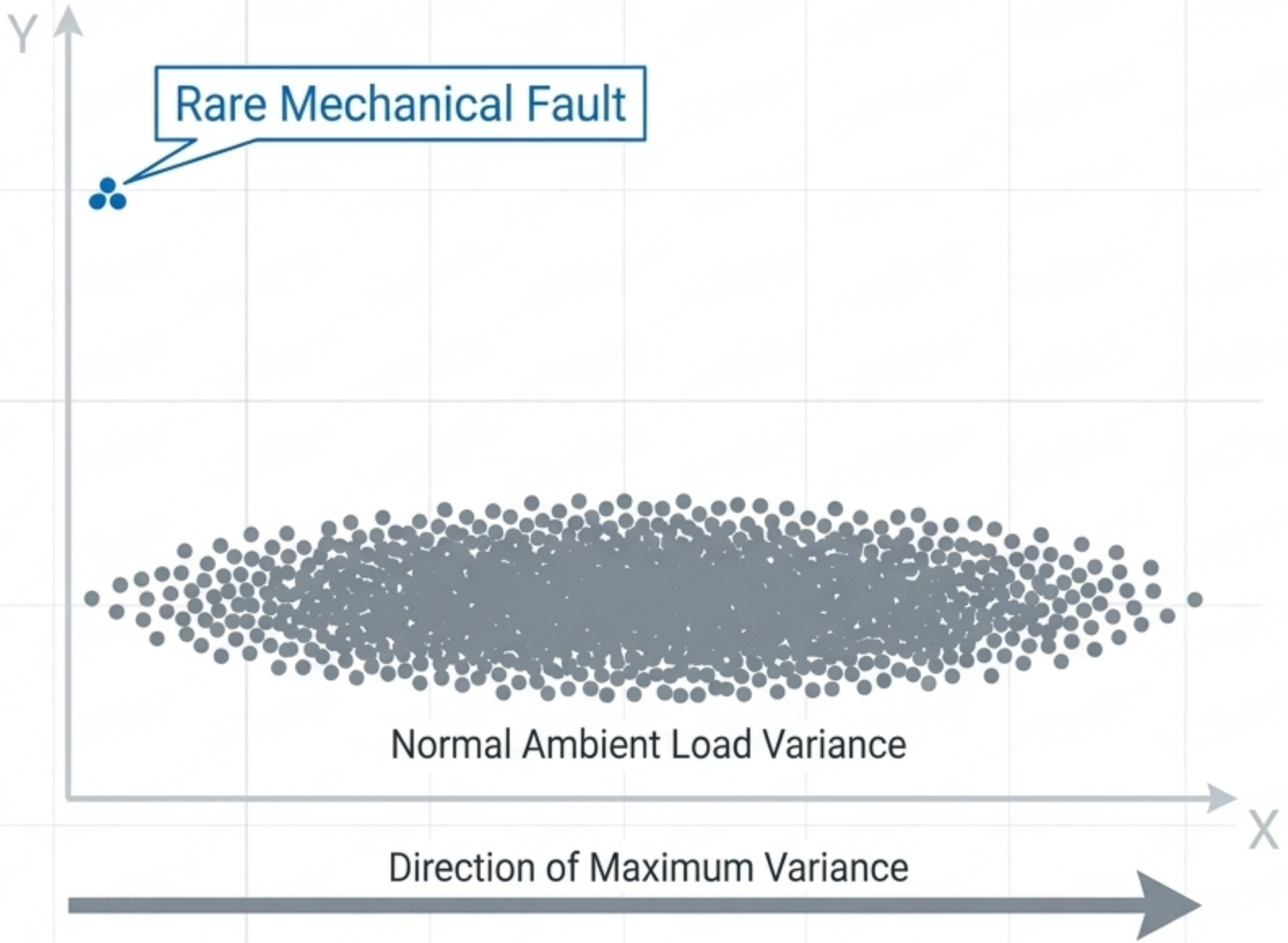
- Cluster quantities are user-selected algorithmic settings.
- A cluster count does not represent proven fault numbers.
- Mathematical grouping requires subsequent engineering interpretation.
- Clusters do not guarantee distinct physical fault classes.



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PCA Variance Versus Task Relevance

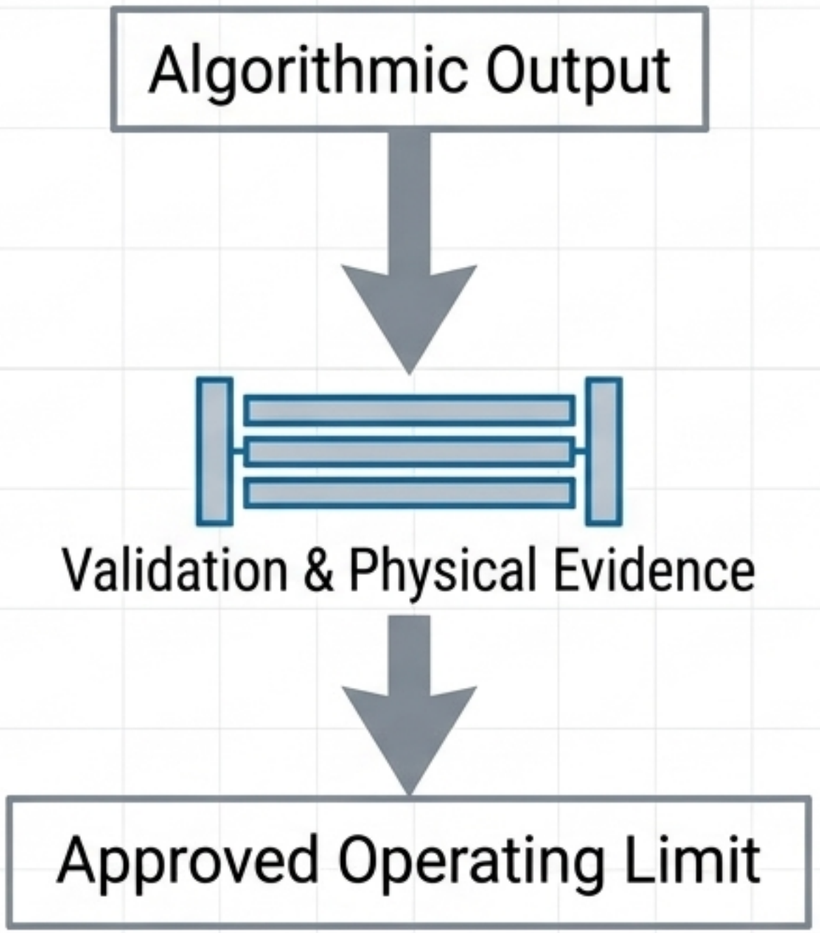
- PCA strictly prioritizes directions of maximum mathematical variance.
- High variation routinely originates from normal ambient load.
- Rare mechanical faults frequently occupy low-variance directions.
- PCA does not filter components by actual fault relevance.



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Justifying Analytical Models

- Algorithmic variance metrics do not equal diagnostic relevance.
- Learned mathematical boundaries are not approved operating limits.
- Complex analytical methods must earn their computational cost.
- Justify all models using task-relevant validation evidence.



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Key Points

1.	k-means mathematically groups points; PCA projects data onto high-variance directions.
2.	Distance-based algorithms demand precise feature scaling reflecting physical units.
3.	Isolate data preprocessing entirely within internal training folds during validation.
4.	High algorithmic variance and clusters do not guarantee mechanical fault relevance.

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